

Tuesday

8

Time: 20 min

Surname:

Initials:

Student number:

Group:

M-number

Use a pen to complete the test. Show all your steps. Use interval notation, where applicable. You are not allowed to use a calculator. The test is printed on both pages.

QUESTION 1 [2]

Can a function be continuous at $x = -3$ but not differentiable at $x = -3$? If your answer is "Yes", prove the statement and if your answer is "No", provide a counter-example.

yes. $f(x) = |x+3|$. f is continuous at $x = -3$.

$$\lim_{x \rightarrow -3^-} \frac{f(x) - f(-3)}{x+3} = \lim_{x \rightarrow -3^-} \frac{|x+3| - 0}{x+3} = \lim_{x \rightarrow -3^-} \frac{-(x+3)}{x+3} = -1$$

$$\lim_{x \rightarrow -3^+} \frac{f(x) - f(-3)}{x+3} = \lim_{x \rightarrow -3^+} \frac{|x+3| - 0}{x+3} = \lim_{x \rightarrow -3^+} \frac{x+3}{x+3} = 1$$

$\Rightarrow f'(-3)$ does not exist $\Rightarrow f$ not differentiable at $x = -3$.

QUESTION 2 [2]

Use the definition of the derivative to find $f'(a)$, where $f(x) = \sqrt{x+1}$ and $a = 1$.

$$f'(1) = \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{1+h+1} - \sqrt{2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{h+2} - \sqrt{2}}{h} \left(\frac{\sqrt{h+2} + \sqrt{2}}{\sqrt{h+2} + \sqrt{2}} \right)$$

$$= \lim_{h \rightarrow 0} \frac{h+2 - 2}{h(\sqrt{h+2} + \sqrt{2})}$$

$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{h+2} + \sqrt{2}} = \frac{1}{2\sqrt{2}}.$$

QUESTION 3 [1, 1.5, 1.5]

Find the derivative of the following functions. Do not simplify your answer.

(i) $f(t) = e^{3-t^2}$

$$f'(t) = e^{3-t^2} \times \frac{d}{dt}(3-t^2)$$

$$= e^{3-t^2}(-2t)$$

(ii) $f(t) = \cos^2(7t) = (\cos 7t)^2$

$$f'(t) = 2(\cos 7t)(-\sin 7t)(7).$$

(iii) $f(x) = \frac{x^2}{1+\sin x}$

$$f'(x) = \frac{\left(\frac{d}{dx} x^2\right)(1+\sin x) - x^2 \frac{d}{dx}(1+\sin x)}{(1+\sin x)^2}$$

$$= \frac{2x(1+\sin x) - x^2(\cos x)}{(1+\sin x)^2}$$